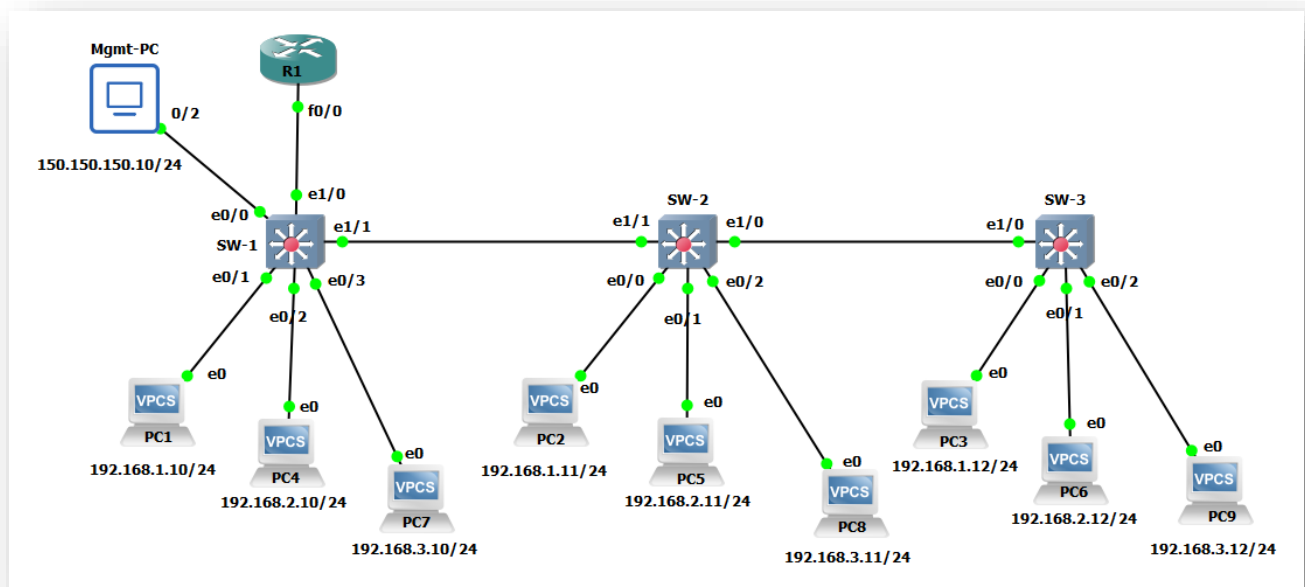


Inter-VLAN Routing (Router on Stick)



Task

1. Configure the SW-1 with IP 150.150.150.1, SW-2 with IP 150.150.150.2 and SW-3 with IP 150.150.150.3 for device management and configure enable password as **ccna** and device access password as **cisco**.
2. Configure IP address on VPCs
3. Configure trunk links between the switches
4. Configure VTP version 3 with domain name rsatechforum.in on all switches. SW-1 should be VTP server and SW-2 & SW-3 as VTP client.
5. Create VLAN-10 as Sale, VLAN-20 as Production and VLAN-30 as Marketing.
6. Assign switchports in respective VLAN as per the topology
7. Test the connectivity between VPCs in same & different VLAN.
8. Configure router R1 for Inter-VLAN routing
9. Test the connectivity between VPCs in different VLAN

For Task-1 to Task-6, refer Lab-27

Note: Mgmt-PC is a cloud which bridge the switch SW-1 with loopback adapter on real PC.

✓ Verification & Testing:

SW-1#sh int trunk

Port	Mode	Encapsulation	Status	Native vlan
Et1/0	on	802.1q	trunking	1
Et1/1	on	802.1q	trunking	1

Port Vans allowed on trunk
Et1/0 1-4094
Et1/1 1-4094

Port Vans allowed and active in management domain
Et1/0 1
Et1/1 1

Port Vans in spanning tree forwarding state and not pruned
Et1/0 1
Et1/1 1

SW-2#sh int trunk

Port	Mode	Encapsulation	Status	Native vlan
Et1/0	on	802.1q	trunking	1
Et1/1	auto	n-802.1q	trunking	1

Port Vans allowed on trunk
Et1/0 1-4094
Et1/1 1-4094

Port Vans allowed and active in management domain
Et1/0 1
Et1/1 1

Port Vans in spanning tree forwarding state and not pruned
Et1/0 1
Et1/1 1

SW-3#sh int trunk

Port	Mode	Encapsulation	Status	Native vlan
Et1/0	auto	n-802.1q	trunking	1

Port Vans allowed on trunk
Et1/0 1-4094

Port Vans allowed and active in management domain
Et1/0 1

Port Vans in spanning tree forwarding state and not pruned
Et1/0 1

✓ Verification & Testing:

SW-1#sh vtp status

VTP Version capable : 1 to 3
VTP version running : 3
VTP Domain Name : rsatechforum.in
VTP Pruning Mode : Disabled
VTP Traps Generation : Disabled
Device ID : aabb.cc80.0100

Feature VLAN:

VTP Operating Mode : Primary Server
Number of existing VLANs : 8
Number of existing extended VLANs : 0
Maximum VLANs supported locally : 4096
Configuration Revision : 2
Primary ID : aabb.cc80.0100
Primary Description : SW-1
MD5 digest : 0x02 0xDF 0x4F 0x3A 0xD1 0x21 0x0E 0x67
0x29 0x74 0x10 0x24 0x33 0x0A 0x31 0xC4

Feature MST:

VTP Operating Mode : Transparent

Feature UNKNOWN:

VTP Operating Mode : Transparent

SW-2#sh vtp status

VTP Version capable : 1 to 3
VTP version running : 3
VTP Domain Name : rsatechforum.in
VTP Pruning Mode : Disabled
VTP Traps Generation : Disabled
Device ID : aabb.cc80.0200

Feature VLAN:

VTP Operating Mode : Client
Number of existing VLANs : 8
Number of existing extended VLANs : 0
Maximum VLANs supported locally : 4096
Configuration Revision : 2
Primary ID : aabb.cc80.0100
Primary Description : SW-1
MD5 digest : 0x02 0xDF 0x4F 0x3A 0xD1 0x21 0x0E 0x67
0x29 0x74 0x10 0x24 0x33 0x0A 0x31 0xC4

Feature MST:

VTP Operating Mode : Transparent

Feature UNKNOWN:

VTP Operating Mode : Transparent

SW-3#sh vtp status

VTP Version capable : 1 to 3
VTP version running : 3
VTP Domain Name : rsatechforum.in

VTP Pruning Mode : Disabled
 VTP Traps Generation : Disabled
 Device ID : aabb.cc80.0300

Feature VLAN:

VTP Operating Mode : Client
 Number of existing VLANs : 8
 Number of existing extended VLANs : 0
 Maximum VLANs supported locally : 4096
 Configuration Revision : 2
 Primary ID : aabb.cc80.0100
 Primary Description : SW-1
 MD5 digest : 0x02 0xDF 0x4F 0x3A 0xD1 0x21 0x0E 0x67
 0x29 0x74 0x10 0x24 0x33 0x0A 0x31 0xC4

Feature MST:

VTP Operating Mode : Transparent

Feature UNKNOWN:

VTP Operating Mode : Transparent

SW-1#sh vlan brief

VLAN Name	Status	Ports
1 default	active	Et0/0, Et1/0, Et1/2, Et1/3 Et2/0, Et2/1, Et2/2, Et2/3 Et3/0, Et3/1, Et3/2, Et3/3
10 Sale	active	Et0/1
20 Production	active	Et0/2
30 Marketing	active	Et0/3
1002 fddi-default	act/unsup	
1003 trcrf-default	act/unsup	
1004 fddinet-default	act/unsup	
1005 trbrf-default	act/unsup	

SW-1#sh vlan summary

Number of existing VLANs : 8
 Number of existing VTP VLANs : 8
 Number of existing extended VTP VLANs : 0

SW-2#sh vlan brief

VLAN Name	Status	Ports
1 default	active	Et0/3, Et1/2, Et1/3, Et2/0 Et2/1, Et2/2, Et2/3, Et3/0 Et3/1, Et3/2, Et3/3

```

10 Sale          active Et0/0
20 Production    active Et0/1
30 Marketing     active Et0/2
1002 fddi-default act/unsup
1003 trcrf-default act/unsup
1004 fddinet-default act/unsup
1005 trbrf-default act/unsup

```

SW-2#sh vlan summary

```

Number of existing VLANs      : 8
Number of existing VTP VLANs  : 8
Number of existing extended VTP VLANs : 0

```

SW-3#sh vlan brief

VLAN Name	Status	Ports
1 default	active	Et0/3, Et1/1, Et1/2, Et1/3 Et2/0, Et2/1, Et2/2, Et2/3 Et3/0, Et3/1, Et3/2, Et3/3
10 Sale	active	Et0/0
20 Production	active	Et0/1
30 Marketing	active	Et0/2
1002 fddi-default	act/unsup	
1003 trcrf-default	act/unsup	
1004 fddinet-default	act/unsup	
1005 trbrf-default	act/unsup	

SW-3#sh vlan summary

```

Number of existing VLANs      : 8
Number of existing VTP VLANs  : 8
Number of existing extended VTP VLANs : 0

```

Task-7: Check connectivity between PCs in same VLAN and different VLANs**Ping PC-2, PC-3, PC-4 and PC-7 from PC-1****PC1> ping 192.168.1.11**

```

84 bytes from 192.168.1.11 icmp_seq=1 ttl=64 time=3.434 ms
84 bytes from 192.168.1.11 icmp_seq=2 ttl=64 time=3.230 ms
84 bytes from 192.168.1.11 icmp_seq=3 ttl=64 time=1.998 ms
84 bytes from 192.168.1.11 icmp_seq=4 ttl=64 time=3.646 ms
84 bytes from 192.168.1.11 icmp_seq=5 ttl=64 time=3.510 ms

```

PC1> ping 192.168.1.12

```

84 bytes from 192.168.1.12 icmp_seq=1 ttl=64 time=1.675 ms
84 bytes from 192.168.1.12 icmp_seq=2 ttl=64 time=5.777 ms
84 bytes from 192.168.1.12 icmp_seq=3 ttl=64 time=5.776 ms
84 bytes from 192.168.1.12 icmp_seq=4 ttl=64 time=7.706 ms
84 bytes from 192.168.1.12 icmp_seq=5 ttl=64 time=7.451 ms

```

PC1> ping 192.168.2.10

host (192.168.1.1) not reachable

PC1> ping 192.168.3.10

host (192.168.1.1) not reachable

Ping PC-5, PC-6, PC-1 and PC-7 from PC-4

PC4> ping 192.168.2.11

84 bytes from 192.168.2.11 icmp_seq=1 ttl=64 time=5.861 ms

84 bytes from 192.168.2.11 icmp_seq=2 ttl=64 time=5.239 ms

84 bytes from 192.168.2.11 icmp_seq=3 ttl=64 time=5.036 ms

84 bytes from 192.168.2.11 icmp_seq=4 ttl=64 time=4.887 ms

84 bytes from 192.168.2.11 icmp_seq=5 ttl=64 time=3.416 ms

PC4> ping 192.168.2.12

84 bytes from 192.168.2.12 icmp_seq=1 ttl=64 time=1.998 ms

84 bytes from 192.168.2.12 icmp_seq=2 ttl=64 time=2.296 ms

84 bytes from 192.168.2.12 icmp_seq=3 ttl=64 time=2.330 ms

84 bytes from 192.168.2.12 icmp_seq=4 ttl=64 time=2.558 ms

84 bytes from 192.168.2.12 icmp_seq=5 ttl=64 time=2.648 ms

PC4> ping 192.168.1.10

host (192.168.2.1) not reachable

PC4> ping 192.168.3.10

host (192.168.2.1) not reachable

Ping PC-8, PC-9, PC-1 and PC-4 from PC-7

PC7> ping 192.168.3.11

84 bytes from 192.168.3.11 icmp_seq=1 ttl=64 time=1.655 ms

84 bytes from 192.168.3.11 icmp_seq=2 ttl=64 time=3.252 ms

84 bytes from 192.168.3.11 icmp_seq=3 ttl=64 time=5.518 ms

84 bytes from 192.168.3.11 icmp_seq=4 ttl=64 time=5.655 ms

84 bytes from 192.168.3.11 icmp_seq=5 ttl=64 time=4.656 ms

PC7> ping 192.168.3.12

84 bytes from 192.168.3.12 icmp_seq=1 ttl=64 time=6.224 ms

84 bytes from 192.168.3.12 icmp_seq=2 ttl=64 time=5.378 ms

84 bytes from 192.168.3.12 icmp_seq=3 ttl=64 time=4.879 ms

84 bytes from 192.168.3.12 icmp_seq=4 ttl=64 time=4.603 ms

84 bytes from 192.168.3.12 icmp_seq=5 ttl=64 time=3.906 ms

PC7> ping 192.168.1.10

host (192.168.3.1) not reachable

PC7> ping 192.168.2.10

host (192.168.3.1) not reachable

Task-8: Configure router R1 for Inter-VLAN routing**R1 Configuration:**

```

R1#config t
R1(config)#int fa0/0
R1(config-if)#no ip address
R1(config-if)#no shut
R1(config-if)#exit
R1(config)#int fa0/0.10
R1(config-subif)#encapsulation dot1q 10
R1(config-subif)#ip address 192.168.1.1 255.255.255.0
R1(config-subif)#exit
R1(config)#int fa0/0.20
R1(config-subif)#encapsulation dot1q 20
R1(config-subif)#ip address 192.168.2.1 255.255.255.0
R1(config-subif)#exit
R1(config)#int fa0/0.30
R1(config-subif)#encapsulation dot1q 30
R1(config-subif)#ip address 192.168.3.1 255.255.255.0
R1(config-subif)#exit
R1(config)#exit
R1#

```

✓ Verification & Testing:**R1#sh ip int brief | exclude down**

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	unassigned	YES	unset	up	up
FastEthernet0/0.10	192.168.1.1	YES	manual	up	up
FastEthernet0/0.20	192.168.2.1	YES	manual	up	up
FastEthernet0/0.30	192.168.3.1	YES	manual	up	up

Task-9: Check connectivity between PCs in different VLANs**Ping PC-4 and PC-7 from PC-1****PC1> ping 192.168.2.10**

```

84 bytes from 192.168.2.10 icmp_seq=1 ttl=63 time=42.538 ms
84 bytes from 192.168.2.10 icmp_seq=2 ttl=63 time=23.028 ms
84 bytes from 192.168.2.10 icmp_seq=3 ttl=63 time=17.489 ms
84 bytes from 192.168.2.10 icmp_seq=4 ttl=63 time=15.111 ms
84 bytes from 192.168.2.10 icmp_seq=5 ttl=63 time=22.195 ms

```

PC1> ping 192.168.3.10

```

84 bytes from 192.168.3.10 icmp_seq=1 ttl=63 time=28.458 ms
84 bytes from 192.168.3.10 icmp_seq=2 ttl=63 time=19.809 ms
84 bytes from 192.168.3.10 icmp_seq=3 ttl=63 time=21.754 ms
84 bytes from 192.168.3.10 icmp_seq=4 ttl=63 time=19.922 ms
84 bytes from 192.168.3.10 icmp_seq=5 ttl=63 time=19.707 ms

```

Ping PC-1 and PC-7 from PC-4**PC4> ping 192.168.1.10**

84 bytes from 192.168.1.10 icmp_seq=1 ttl=63 time=18.762 ms
 84 bytes from 192.168.1.10 icmp_seq=2 ttl=63 time=17.338 ms
 84 bytes from 192.168.1.10 icmp_seq=3 ttl=63 time=19.922 ms
 84 bytes from 192.168.1.10 icmp_seq=4 ttl=63 time=19.508 ms
 84 bytes from 192.168.1.10 icmp_seq=5 ttl=63 time=14.325 ms

PC4> ping 192.168.3.10

84 bytes from 192.168.3.10 icmp_seq=1 ttl=63 time=24.148 ms
 84 bytes from 192.168.3.10 icmp_seq=2 ttl=63 time=18.743 ms
 84 bytes from 192.168.3.10 icmp_seq=3 ttl=63 time=20.716 ms
 84 bytes from 192.168.3.10 icmp_seq=4 ttl=63 time=20.884 ms
 84 bytes from 192.168.3.10 icmp_seq=5 ttl=63 time=16.725 ms

Ping PC-1 and PC-4 from PC-7**PC7> ping 192.168.1.10**

84 bytes from 192.168.1.10 icmp_seq=1 ttl=63 time=17.141 ms
 84 bytes from 192.168.1.10 icmp_seq=2 ttl=63 time=14.967 ms
 84 bytes from 192.168.1.10 icmp_seq=3 ttl=63 time=19.162 ms
 84 bytes from 192.168.1.10 icmp_seq=4 ttl=63 time=18.740 ms
 84 bytes from 192.168.1.10 icmp_seq=5 ttl=63 time=19.763 ms

PC7> ping 192.168.2.10

84 bytes from 192.168.2.10 icmp_seq=1 ttl=63 time=20.442 ms
 84 bytes from 192.168.2.10 icmp_seq=2 ttl=63 time=16.986 ms
 84 bytes from 192.168.2.10 icmp_seq=3 ttl=63 time=16.751 ms
 84 bytes from 192.168.2.10 icmp_seq=4 ttl=63 time=15.330 ms
 84 bytes from 192.168.2.10 icmp_seq=5 ttl=63 time=17.198 ms

R1#sh int fa0/0 | section address

Hardware is i82543 (Livengood), address is **ca01.1cac.0008** (bia ca01.1cac.0008)
 R1#

PC1> sh ip all

NAME	IP/MASK	GATEWAY	MAC	DNS
PC1	192.168.1.10/24	192.168.1.1	00:50:79:66:68:00	

PC4> sh ip all

NAME	IP/MASK	GATEWAY	MAC	DNS
PC4	192.168.2.10/24	192.168.2.1	00:50:79:66:68:03	

PC7> sh ip all

NAME	IP/MASK	GATEWAY	MAC	DNS
PC7	192.168.3.10/24	192.168.3.1	00:50:79:66:68:06	

R1#sh arp

Protocol	Address	Age (min)	Hardware Addr	Type	Interface
Internet	192.168.1.1	-	ca01.1cac.0008	ARPA	FastEthernet0/0.10
Internet	192.168.1.10	1	0050.7966.6800	ARPA	FastEthernet0/0.10
Internet	192.168.2.1	-	ca01.1cac.0008	ARPA	FastEthernet0/0.20
Internet	192.168.2.10	1	0050.7966.6803	ARPA	FastEthernet0/0.20
Internet	192.168.3.1	-	ca01.1cac.0008	ARPA	FastEthernet0/0.30
Internet	192.168.3.10	1	0050.7966.6806	ARPA	FastEthernet0/0.30

R1#

SW-1#sh mac address-table

Mac Address Table

Vlan	Mac Address	Type	Ports
1	0200.4c4f.4f50	DYNAMIC	Et0/0
1	aabb.cc00.0511	DYNAMIC	Et1/1
1	ca01.1cac.0008	DYNAMIC	Et1/0
10	0050.7966.6800	DYNAMIC	Et0/1
10	ca01.1cac.0008	DYNAMIC	Et1/0
20	0050.7966.6803	DYNAMIC	Et0/2
20	ca01.1cac.0008	DYNAMIC	Et1/0
30	0050.7966.6806	DYNAMIC	Et0/3
30	ca01.1cac.0008	DYNAMIC	Et1/0

Total Mac Addresses for this criterion: 9
SW-1#

SW-2#sh mac address-table

Mac Address Table

Vlan	Mac Address	Type	Ports
1	0200.4c4f.4f50	DYNAMIC	Et1/1
1	aabb.cc00.0411	DYNAMIC	Et1/1
1	aabb.cc00.0601	DYNAMIC	Et1/0
1	ca01.1cac.0008	DYNAMIC	Et1/1
10	0050.7966.6800	DYNAMIC	Et1/1
10	aabb.cc00.0411	DYNAMIC	Et1/1
20	aabb.cc00.0411	DYNAMIC	Et1/1
20	ca01.1cac.0008	DYNAMIC	Et1/1
30	aabb.cc00.0411	DYNAMIC	Et1/1
30	ca01.1cac.0008	DYNAMIC	Et1/1

Total Mac Addresses for this criterion: 10
SW-2#

SW-3#sh mac address-table

Mac Address Table

Vlan	Mac Address	Type	Ports
1	0200.4c4f.4f50	DYNAMIC	Et1/0
1	aabb.cc00.0501	DYNAMIC	Et1/0
1	ca01.1cac.0008	DYNAMIC	Et1/0
10	0050.7966.6800	DYNAMIC	Et1/0
10	aabb.cc00.0501	DYNAMIC	Et1/0
20	aabb.cc00.0501	DYNAMIC	Et1/0
20	ca01.1cac.0008	DYNAMIC	Et1/0
30	aabb.cc00.0501	DYNAMIC	Et1/0
30	ca01.1cac.0008	DYNAMIC	Et1/0

Total Mac Addresses for this criterion: 9

SW-3#

Important Commands:

Sh vlan brief
Sh vlan summary
Sh vtp status
Sh arp
Sh ip int brief | exclude down
Sh mac address-table

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